

OCR Further Mathematics
Statistics - Continuous Random Variables
Question Paper
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

1(a) The continuous random variable Y has a uniform distribution on $[0, 2]$.

It is given that $E[a \cos(aY)] = 0.3$, where a is a constant between 0 and 1, and aY is measured in radians.

Determine the value of the constant a .

[5]

(b) Determine the 60th percentile of Y^2 .

[6]

2(a) The continuous random variable X has probability density function

$$f(x) = \begin{cases} \frac{k}{x^n} & x \geq 1, \\ 0 & \text{otherwise,} \end{cases}$$

where n and k are constants and n is an integer greater than 1.

Find k in terms of n .

[3]

(b) (i) When $n = 4$, find the cumulative distribution function of X .

[3]

(ii) Hence determine $P(X > 7 \mid X > 5)$ when $n = 4$.

[4]



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(c) Determine the values of n for which $\text{Var}(X)$ is not defined.

[5]



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3(a) The numbers of CD players sold in a shop on three consecutive weekends were 7, 6 and 2. It may be assumed that sales of CD players occur randomly and that nobody buys more than one CD player at a time. The number of CD players sold on a randomly chosen weekend is denoted by X .

How appropriate is the Poisson distribution as a model for X ?

[2]

(b) Now assume that a Poisson distribution with mean 5 is an appropriate model for X . Find

(i) $P(X = 6)$,

[2]

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(ii) $P(X \geq 8)$.

[2]

- (c) The number of integrated sound systems sold in a weekend at the same shop can be assumed to have the distribution $Po(7.2)$.

Find the probability that on a randomly chosen weekend the total number of CD players and integrated sound systems sold is between 10 and 15 inclusive.

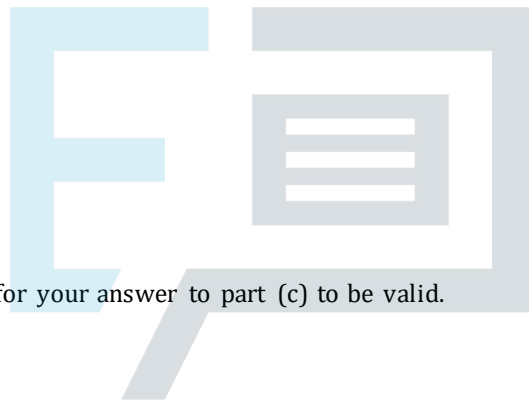
[3]



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(d) State an assumption needed for your answer to part (c) to be valid.

[1]

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(e) Give a reason why the assumption in part (d) may not be valid in practice.

[1]

- 4 The continuous random variable X has the distribution $N(\mu, 30)$. The mean of a random sample of 8 observations of X is 53.1.

Determine a 95% confidence interval for μ . You should give the end points of the interval correct to 4 significant figures.

[4]



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5(a) The continuous random variable T has cumulative distribution function

$$F(t) = \begin{cases} 0 & t < 0, \\ 1 - e^{-0.25t} & t \geq 0. \end{cases}$$

Find the cumulative distribution function of $2T$.

[3]

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(b)

Show that, for constant k , $E(e^{kt}) = \frac{1}{1-4k}$.

You should state with a reason the range of values of k for which this result is valid.

[7]



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(c) T is the time before a certain event occurs.

Show that the probability that no event occurs between time $T = 0$ and time $T = \theta$ is the same as the probability that the value of a random variable with the distribution $Po(\lambda)$ is 0, for a certain value of λ . You should state this value of λ in terms of θ .

[4]



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- 6 A continuous random variable X has probability density function given by the following function, where a is a constant.

$$f(x) = \begin{cases} \frac{2x}{a^2} & 0 \leq x \leq a, \\ 0 & \text{otherwise.} \end{cases}$$

The expected value of X is 4.

- (a) Show that $a = 6$.

[3]

Five independent observations of X are obtained, and the largest of them is denoted by M .

- (b) Find the cumulative distribution function of M .

[8]

- 7 A survey is carried out into the length of time for which customers wait for a response on a telephone helpline. A statistician who is analysing the results of the survey starts by modelling the waiting time, x minutes, by an exponential distribution with probability density function (PDF)

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \\ 0 & x < 0. \end{cases}$$

- (a) In this question you must show detailed reasoning.

The mean waiting time is found to be 5.0 minutes. Show that $\lambda = 0.2$.

[4]



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- (b) Use the model to calculate the probability that a customer has to wait longer than 20 minutes for a response.

[2]

In practice it is found that no customer waits for more than 15 minutes for a response. The statistician constructs an improved model to incorporate this fact.

(c) On the diagram below, sketch the following, labelling the curves clearly:

(i) the PDF of the model using the exponential distribution,

[1]

(ii) a possible PDF for the improved model.

[2]



8 The function $f(x)$ is defined by

$$f(x) = \begin{cases} \frac{1}{4}x(4-x^2) & 0 \leq x \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Show that $f(x)$ satisfies the conditions for a probability density function.

[3]

(b) Find the value of a such that $P(X < a) = \frac{15}{16}$.

[6]

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- 9 A spinner has 5 edges. Each edge is numbered with a different integer from 1 to 5. When the spinner is spun, it is equally likely to come to rest on any one of the edges. The spinner is spun 100 times. The number of times on which the spinner comes to rest on the edge numbered 5 is denoted by X .

(a) Write down

(i) $E(X)$, [1]

(ii) $\text{Var}(X)$. [1]

- (b) Use a normal distribution with the same mean and variance as in your answers to part (a) to estimate the smallest value of n such that $P(X \geq n) < 0.02$. [3]

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- (c) Use the binomial distribution to find exactly the smallest value of n such that $P(X \geq n) < 0.02$. Show the values of all relevant calculations. [3]

10 A continuous random variable X has cumulative distribution function given by

$$F(x) = \begin{cases} 1 - \frac{1}{x^3} & x \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(\sqrt{X})$.

[5]



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11 A continuous random variable X has probability density function

$$f(x) = \begin{cases} \frac{1}{64}x(16-x^2) & 0 \leq x \leq 4, \\ 0 & \text{otherwise.} \end{cases}$$

(i) Find the value of $E(X)$.

[3]



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(ii) The upper quartile q is defined by the condition $P(X \leq q) = \frac{3}{4}$.

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(a) Show that q satisfies the equation

$$q^4 - 32q^2 + 192 = 0.$$

[3]

(b) Hence find the exact value of q .

[3]



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- 12 The editor of a scientific journal receives articles, for possible publication, at random times throughout the year.
- (i) State two conditions needed for the number of articles received in one randomly chosen week to be well modelled by a Poisson distribution. [2]

Assume now that the number of articles received in one week can be modelled by the distribution $Po(2.4)$.

- (ii) Find the probability that in a randomly chosen 2-week period fewer than 5 articles are received. [2]

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- (iii) The number of articles received in one randomly chosen week is denoted by R . Given that $P(R = r) = 2.5 \times P(R = r + 1)$, use an algebraic method to find the value of r . [4]

- (iv) Use an appropriate approximation to find the probability that in a randomly chosen 50-week period at least 140 articles are received. [5]



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- 13 One game of roulette consists of throwing a ball onto a spinning wheel, which has 37 slots, numbered 0 to 36. On each throw the ball is equally likely to come to rest in any one of the 37 slots on the wheel, independently of all other throws. The number of the slot in which the ball comes to rest is the score of the game.
- (i) Use a suitable approximation to find the probability that on 74 throws of the ball a score of 0 is obtained at least four times. Justify your approximation. [4]



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- (ii) 18 of the 37 possible scores are called “impair”. 148 games are played. The probability that impair is obtained at least N times is less than 0.0025. Use a suitable approximation to find the smallest possible value of N . [6]



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- 14 A continuous random variable W has the distribution $N(\mu, \sigma^2)$. It is given that $P(W < 70.0) = 0.8$ and $P(W > 81.0) = 0.1$. Find the value of a if $P(W < a) = 0.1$.

[7]



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- 15 The object distance U and the image distance V for a convex lens are variables. They are related to the focal length f of the lens, which is a constant, by the formula

$$\frac{1}{U} + \frac{1}{V} = \frac{1}{f}.$$

U is a random variable with a continuous uniform distribution over the interval from $3f$ to $4f$. Find the probability density function of V .

[6]



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16 A continuous random variable X has probability density function

$$f(x) = \begin{cases} kx^2 & 0 \leq x \leq 2, \\ kx^2 - 1.5x + 3 & 2 < x \leq 4, \\ 0 & \text{otherwise,} \end{cases}$$

where k is a constant.

(i) Show that $k = \frac{3}{16}$.

[2]



(ii) Find $P(1.5 \leq X \leq 3)$.

[3]

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(iii) Show that the upper quartile of X is 2.41, correct to 2 decimal places.

[4]



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17 The continuous random variable X has cumulative distribution function given by

$$F(x) = \begin{cases} 0 & x < 0, \\ \frac{1}{16}x^2 & 0 \leq x \leq 4, \\ 1 & x > 4. \end{cases}$$

(a) The random variable Y is defined by $Y = \frac{1}{X^2}$. Find the cumulative distribution function of Y . [5]

(b) Show that $E(Y)$ is not defined. [4]

18 A continuous random variable X has probability density function given by

$$f(x) = \begin{cases} 0.8e^{-0.8x} & x \geq 0, \\ 0 & x < 0. \end{cases}$$

(a) Find the mean and variance of X .

[4]

The lifetime of a certain organism is thought to have the same distribution as X . The lifetimes in days of a random sample of 60 specimens of the organism were found. The observed frequencies, together with the expected frequencies correct to 3 decimal places, are given in the table.

Range	$0 \leq x < 1$	$1 \leq x < 2$	$2 \leq x < 3$	$3 \leq x < 4$	$x \geq 4$
Observed	24	22	10	3	1
Expected	33.040	14.846	6.671	2.997	2.446

(b) Show how the expected frequency for $1 \leq x < 2$ is obtained.

[4]

(c) Carry out a goodness of fit test at the 5% significance level.

[7]



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19 The mass J kg of a bag of randomly chosen Jersey potatoes is a normally distributed random variable with mean 1.00 and standard deviation 0.06. The mass K kg of a bag of randomly chosen King Edward potatoes is an independent normally distributed random variable with mean 0.80 and standard deviation 0.04.

- (a) Find the probability that the total mass of 6 bags of Jersey potatoes and 8 bags of King Edward potatoes is greater than 12.70 kg. [3]

- (b) Find the probability that the mass of one bag of King Edward potatoes is more than 75% of the mass of one bag of Jersey potatoes. [3]

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20 A continuous random variable X has probability density function

$$f(x) = \begin{cases} ax^3 & 0 \leq x \leq 1, \\ ax^2 & 1 < x \leq 2, \\ 0 & \text{otherwise,} \end{cases}$$

where a is a constant.

(i) Show that $a = \frac{12}{31}$.

[2]

(ii) Find $E(X)$.

[2]



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It is thought that the time taken by a student to complete a task can be well modelled by X . The times taken by 992 randomly chosen students are summarised in the table, together with some of the expected frequencies.

Time	$0 \leq x < 0.5$	$0.5 \leq x < 1$	$1 \leq x < 1.5$	$1.5 \leq x \leq 2$
Observed frequency	8	92	279	613
Expected frequency	6	90		

- (iii) Find the other expected frequencies and test, at the 5% level of significance, whether the data can be well modelled by X .

[8]

- 21 The radius, R , of a sphere is a random variable with a continuous uniform distribution between 0 and 10.

- (i) Find the cumulative distribution function and probability density function of A , the surface area of the sphere.

[8]

- (ii) Find $P(A \leq 200\pi)$.

[2]

22 A continuous random variable X has probability density function

$$f(x) = \begin{cases} kx & 0 \leq x < 2, \\ \frac{k(4-x)^2}{2} & 2 \leq x \leq 4, \\ 0 & \text{otherwise,} \end{cases}$$

where k is a constant.

(i) Show that $k = \frac{3}{10}$.

[3]

(ii) Find $E(X)$.

[4]

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(iii) Find the cumulative distribution function of X .

[4]



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(iv) Find the upper quartile of X , correct to 3 significant figures.

[3]

23 The continuous random variable X has probability density function given by

$$f(x) = \begin{cases} k \sin x & 0 \leq x \leq \frac{1}{2}\pi, \\ k(2 - \frac{2x}{\pi}) & \frac{1}{2}\pi \leq x \leq \pi, \\ 0 & \text{otherwise,} \end{cases}$$

where k is a constant.

(i) Show that $k = \frac{4}{4 + \pi}$.

[4]



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(ii) Find $E(X)$, correct to 3 significant figures, showing all necessary working.

[4]

24 A rectangle of area $A \text{ m}^2$ has a perimeter of 20 m and each of the two shorter sides are of length $X \text{ m}$, where X is uniformly distributed between 0 and 2.

(i) Write down an expression for A in terms of X , and hence show that $A = 25 - (X - 5)^2$.

[3]



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(ii) Write down the probability density function of X .

[1]



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(iii) Show that the cumulative distribution function of A is

$$F(a) = \begin{cases} 0 & a < 0, \\ \frac{1}{2}(5 - \sqrt{25 - a}) & 0 \leq a \leq 16, \\ 1 & a > 16. \end{cases}$$

[5]

(iv) Find the probability density function of A .

[2]

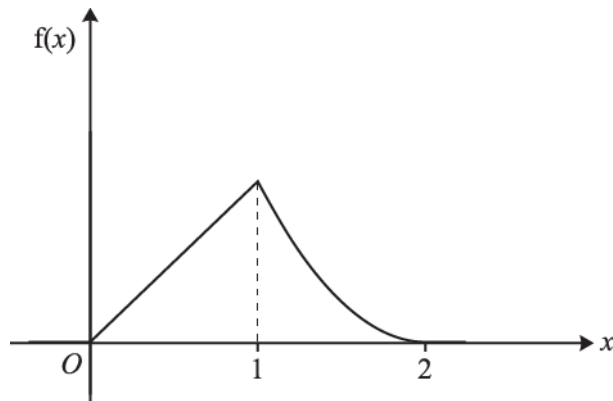


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25



The continuous random variable X has probability density function given by

$$f(x) = \begin{cases} ax & 0 < x \leq 1, \\ b(2-x)^2 & 1 < x \leq 2, \\ 0 & \text{otherwise,} \end{cases}$$

where a and b are constants. The graph is shown in the above diagram.

(i) Find the values of a and b .

[5]

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(ii) Find the value of $E\left(\frac{1}{X}\right)$.

[3]

26 The continuous random variable Y has probability density function given by

$$f(y) = \begin{cases} \ln(y) & 1 \leq y \leq e, \\ 0 & \text{otherwise} \end{cases}$$

(i) Verify that this is a valid probability density function.

[4]

(ii) Show that the (cumulative) distribution function of Y is given by

$$F(y) = \begin{cases} 0 & y < 1, \\ y \ln y - y + 1 & 1 \leq y \leq e \\ 1 & \text{otherwise} \end{cases}$$

[3]

(iii) Verify that the upper quartile of Y lies in the interval $[2.45, 2.46]$.

[2]

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(iv) Find the (cumulative) distribution function of X where $X = \ln Y$.

[4]



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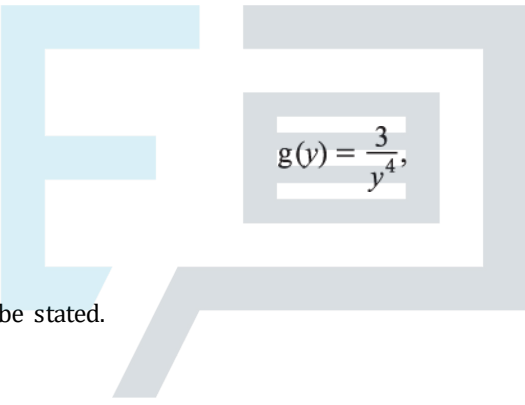
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27 The continuous random variable X has probability density function given by

$$f(x) = \begin{cases} \frac{3}{2}\sqrt{x} & 0 < x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

The random variable Y is given by $Y = \frac{1}{\sqrt{X}}$.

- (i) Find the (cumulative) distribution function of Y , and hence show that its probability density function is given by


$$g(y) = \frac{3}{y^4},$$

for a set of values of y to be stated.

[7]

- (ii) Find the value of $E(Y^2)$.

[2]

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28 A continuous random variable X has probability density function

$$f(x) = \begin{cases} ax^{-3} + bx^{-4} & x \geq 1, \\ 0 & \text{otherwise,} \end{cases}$$

where a and b are constants.

(i) Explain what the letter x represents.

[1]



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It is given that $P(X > 2) = \frac{3}{16}$.

(ii) Show that $a = 1$, and find the value of b .

[7]



Exam Papers Practice [3]

(iii) Find $E(X)$.

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29 A continuous random variable X has probability density function

$$f(x) = \begin{cases} \frac{3}{2a^3} x^2 & -a \leq x \leq a, \\ 0 & \text{otherwise,} \end{cases}$$

where a is a constant.

(i) It is given that $P(-3 \leq X \leq 3) = 0.125$. Find the value of a in this case.

[4]



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(ii) It is given instead that $\text{Var}(X) = 1.35$. Find the value of a in this case.

[5]



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(iii) Explain the relationship between x and X in this question.

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[1]

30 A continuous random variable X has probability density function

$$f(x) = \begin{cases} \frac{1}{2}\pi \sin(\pi x) & 0 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

(i) Show that this is a valid probability density function.

[4]



(ii) Sketch the curve $y = f(x)$ and write down the value of $E(X)$.

[3]

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(iii) Find the value q such that $P(X > q) = 0.75$.

[3]



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(iv) Write down an expression, including an integral, for $\text{Var}(X)$. (Do not attempt to evaluate the integral.)

[2]



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- (v) A student states that “ X is more likely to occur when x is close to $E(X)$.” Give an improved version of this statement.

[1]



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31 Two random variables S and T have probability density functions given by

$$f_S(x) = \begin{cases} \frac{3}{a^3}(x-a)^2 & 0 \leq x \leq a, \\ 0 & \text{otherwise,} \end{cases}$$
$$f_T(x) = \begin{cases} c & 0 \leq x \leq a, \\ 0 & \text{otherwise,} \end{cases}$$

where a and c are constants.

(i) On a single diagram sketch both probability density functions.

[3]



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(ii) Calculate the mean of S , in terms of a .

[5]



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(iii) Use your diagram to explain which of S or T has the bigger variance. (Answers obtained by calculation will score no marks.)

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[2]

- 32 The random variable X denotes the yield, in kilograms per acre, of a certain crop. Under the standard treatment it is known that $E(X) = 38.4$. Under a new treatment, the yields of 50 randomly chosen regions can be summarised as

$$n = 50, \quad \Sigma x = 1834.0, \quad \Sigma x^2 = 70027.37.$$

Test at the 1% level whether there has been a change in the mean crop yield.

[11]



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33 A continuous random variable X has probability density function

$$f(x) = \begin{cases} kx & 0 \leq x \leq a, \\ 0 & \text{otherwise,} \end{cases}$$

where k and a are constants.

(i) State what the letter x represents.

[1]

(ii) Find k in terms of a .

[2]

(iii) Find $\text{Var}(X)$ in terms of a .

[6]

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